Project Requirement

Group 4

Project title: **ម៉ាប់ដូង – Mlup Dong (Restaurant Management System)**	**Last updated:** Feb 20, 2026
Project manager: Sophon Sophea	**Project status:** ❌ Not started

1.Project overview

Mlup Dong (Restaurant Management System) is a smart restaurant system that allows customers to order food and drink. by scanning the QR code at their table. Customers do not need to call staff, and they can browse the menu, place orders, and track preparation time directly from their smartphone.

The system also helps restaurant owners and staff manage daily operations, including menu management, staff management, sales analytics, and kitchen workflow

2. Objectives

. Reduce waiting time for customers

· Reduce workload of waiters

· Provide modern dining experience (QR ordering)

· Improve kitchen workflow efficiency

· Provide business insights for the restaurant owner

3. User Role

Role	Description
Customer	Scan QR code, place food orders, and track waiting time
Chef	View incoming orders and update cooking status
Staff/Waiter	Manage tables and assist customers
Admin/Owner	Manage menu, staff, reports, and analytics

4. Main Features

4.1 Customer Features

· Scan the QR code at the table to open the website.

· View food & drink menu

· Filter by category (food / drink / promotion)

· Add items to cart

· Place order

· See order status (Pending / Cooking / Ready / Served)

· Live waiting time counter

· See order history

4.2 Chef (Kitchen Display System)

· View new orders in real-time

· See table number & ordered items

· Confirm Start Cooking

· Confirm Ready

· Cooking timer auto-calculate waiting time

· Priority queue (first order first cook)

4.3 Admin/Owner Dashboard

· Menu Management (add, update, images, price, availability, categories)

· Staff Management (add, update, remove, assign roles)

· Table Management (create tables, generate QR codes)

· Sales & Reports (daily, monthly, yearly income)

· Best-selling food/drink chart

· Order statistics graph

· Peak hour analysis

4.4 Waiter Feature

Assist customer orders

· Change order status to served

· Transfer table (optional)

· Cancel order (permission based)

5. System Modules

· Authentication & Authorization

· QR Table Ordering System

· Order Management System

· Kitchen Display System

· Menu Management

· Staff Management

· Reporting & Analytics Dashboard

· Notification & Real-time Update (Redis WebSocket

6. Database (Main Tables)

No	Table Name	Description
1	users	Store login information (admin, staff, chef)
2	roles	Store user roles (Admin, Chef, Waiter)
3	tables	Store restaurant table information (table number, QR code)
4	categories	Store food & drink categories
5	menu_items	Store food & drink details (name, price, image, availability)
6	Order	Store customer order information
7	order_items	Store items inside each order
8	Payment	Store items inside each order
9	staff	Store staff details
10	order status_logs	Store order status history (Pending, Cooking, Ready, Served)

7. Workflow

1. Customer Flow

1. Scan QR
2. Open menu website
3. Select food
4. Place order
5. Track waiting timer
6. Receive food

2. Kitchen Flow

7. Receive order

8. Confirm cooking

9. Prepare food

10. Mark ready

3. Owner Flow

11. Monitor dashboard

12. View sales

13. Analyze best sellers

14. Manage menu & staf

8. Non-Functional Requirements

· Real-time updates using Redis

· Responsive mobile design

· Fast loading menu (< 2 seconds)

· Secure authentication (JWT / Session)

· Support at least 100+ concurrent users

· Automatic order synchronization (no refresh needed)

9. Expected Outcome

· Replace traditional paper menu

· Reduce order mistakes

· Speed up kitchen operations

· Provide business analytics

· Improve customer experience

10. Future Improvements

· Online payment integration (ABA / KHQR)

· Promotion & coupon system

· Customer loyalty points

· Multi-branch restaurant support